

Project

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```
#libraries
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.4.4      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(car)

## Warning: package 'car' was built under R version 4.3.3
## Loading required package: carData
## Warning: package 'carData' was built under R version 4.3.3
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some

library(lmtest)

## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

library(MASS)

##
```

```

## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##      select
library(ggplot2)

nba_data <- read_csv("Data/nba_team_stats_00_to_23.csv")

## Rows: 716 Columns: 29
## -- Column specification -----
## Delimiter: ","
## chr  (2): Team, season
## dbl  (27): teamstatspk, games_played, wins, losses, win_percentage, Min, poin...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
#Create ranking by wins (descending)

nba_data <- nba_data %>%
  group_by(season) %>%
  mutate(ranking = rank(-wins, ties.method = "first")) %>%
  arrange(season, ranking)

write_csv(nba_data, "Data/nba_team_stats_ranked.csv")

nba_model <- read_csv("Data/nba_team_stats_ranked.csv")

## Rows: 716 Columns: 30
## -- Column specification -----
## Delimiter: ","
## chr  (2): Team, season
## dbl  (28): teamstatspk, games_played, wins, losses, win_percentage, Min, poin...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
# for reproducibility
set.seed(325)
n <- nrow(nba_model)
train_idx <- sample(seq_len(n), size = floor(0.7 * n))

nba_train <- nba_model[train_idx, ]
nba_test  <- nba_model[-train_idx, ]

vars_use <- c(
  "wins",
  "field_goal_percentage",
  "three_point_percentage",
  "free_throw_percentage",
  "assists",
  "turnovers",
  "offensive_rebounds",
  "defensive_rebounds",

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    "steals",
    "blocks",
    "personal_fouls",
    "personal_fouls_drawn"
  )

  # Compute mean and SD for each variable
  summary_stats <- nba_model %>%
    dplyr::select(all_of(vars_use)) %>%
    dplyr::summarise(across(everything(),
                           list(mean = mean, sd = sd),
                           na.rm = TRUE))

## Warning: There was 1 warning in `dplyr::summarise()`.
## i In argument: `across(everything(), list(mean = mean, sd = sd), na.rm =
##   TRUE)`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
## # Previously
##   across(a:b, mean, na.rm = TRUE)
##
## # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))

summary_stats

## # A tibble: 1 x 24
##   wins_mean wins_sd field_goal_percentage_mean field_goal_percentage_sd
##   <dbl> <dbl> <dbl> <dbl>
## 1    40.2  12.2    45.6    1.69
## # i 20 more variables: three_point_percentage_mean <dbl>,
## #   three_point_percentage_sd <dbl>, free_throw_percentage_mean <dbl>,
## #   free_throw_percentage_sd <dbl>, assists_mean <dbl>, assists_sd <dbl>,
## #   turnovers_mean <dbl>, turnovers_sd <dbl>, offensive_rebounds_mean <dbl>,
## #   offensive_rebounds_sd <dbl>, defensive_rebounds_mean <dbl>,
## #   defensive_rebounds_sd <dbl>, steals_mean <dbl>, steals_sd <dbl>,
## #   blocks_mean <dbl>, blocks_sd <dbl>, personal_fouls_mean <dbl>, ...

#full model
full_mod <- lm(wins ~ field_goal_percentage +
               three_point_percentage +
               free_throw_percentage +
               assists + turnovers +
               offensive_rebounds +
               defensive_rebounds +
               steals + blocks +
               personal_fouls +
               personal_fouls_drawn,
               data = nba_train)

summary(full_mod)

##

```

```
## Call:
## lm(formula = wins ~ field_goal_percentage + three_point_percentage +
##     free_throw_percentage + assists + turnovers + offensive_rebounds +
##     defensive_rebounds + steals + blocks + personal_fouls + personal_fouls_drawn,
##     data = nba_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.8537  -5.2650   0.3927   4.8536  25.9963
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.236e+02  1.651e+01 -13.545 < 2e-16 ***
## field_goal_percentage  4.049e+00  3.152e-01  12.847 < 2e-16 ***
## three_point_percentage  1.482e+00  2.296e-01   6.455 2.61e-10 ***
## free_throw_percentage -1.387e-01  1.238e-01  -1.120 0.263166
## assists          -1.933e-02  2.537e-03  -7.620 1.32e-13 ***
## turnovers         -4.237e-02  4.264e-03  -9.938 < 2e-16 ***
## offensive_rebounds   2.460e-02  3.902e-03   6.304 6.49e-10 ***
## defensive_rebounds   2.346e-02  2.308e-03  10.165 < 2e-16 ***
## steals            4.660e-02  5.425e-03   8.589 < 2e-16 ***
## blocks            2.054e-02  5.693e-03   3.608 0.000341 ***
## personal_fouls       4.176e-03  2.718e-03   1.536 0.125160
## personal_fouls_drawn -2.951e-03  5.981e-04  -4.934 1.11e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.536 on 489 degrees of freedom
## Multiple R-squared:  0.6095, Adjusted R-squared:  0.6007
## F-statistic: 69.39 on 11 and 489 DF,  p-value: < 2.2e-16
```

```
vif(full_mod)
```

```
## field_goal_percentage three_point_percentage free_throw_percentage
##           2.392643           1.594904           1.217149
##           assists           turnovers           offensive_rebounds
##           2.383574           1.948425           1.876417
##           defensive_rebounds           steals           blocks
##           1.884339           1.332627           1.253295
##           personal_fouls personal_fouls_drawn
##           1.949341           1.457542
```

```
step_mod <- stepAIC(full_mod, direction = "both", trace = FALSE)
summary(step_mod)
```

```
##
## Call:
## lm(formula = wins ~ field_goal_percentage + three_point_percentage +
##     assists + turnovers + offensive_rebounds + defensive_rebounds +
##     steals + blocks + personal_fouls + personal_fouls_drawn,
##     data = nba_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.2202  -5.1584   0.3634   4.8508  26.1955
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.340e+02  1.363e+01 -17.165 < 2e-16 ***
## field_goal_percentage  4.043e+00  3.152e-01  12.825 < 2e-16 ***
## three_point_percentage  1.468e+00  2.294e-01   6.402 3.61e-10 ***
## assists         -1.968e-02  2.518e-03  -7.813 3.42e-14 ***
## turnovers       -4.122e-02  4.138e-03  -9.959 < 2e-16 ***
## offensive_rebounds  2.518e-02  3.869e-03   6.508 1.89e-10 ***
## defensive_rebounds  2.331e-02  2.304e-03  10.114 < 2e-16 ***
## steals          4.687e-02  5.421e-03   8.646 < 2e-16 ***
## blocks          2.087e-02  5.687e-03   3.669 0.00027 ***
## personal_fouls    3.855e-03  2.704e-03   1.426 0.15461
## personal_fouls_drawn -2.895e-03  5.962e-04  -4.856 1.61e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.538 on 490 degrees of freedom
## Multiple R-squared:  0.6085, Adjusted R-squared:  0.6005
## F-statistic: 76.17 on 10 and 490 DF, p-value: < 2.2e-16
```

```
vif(step_mod)
```

```
## field_goal_percentage three_point_percentage assists
##           2.391849           1.590150           2.348091
##           turnovers      offensive_rebounds defensive_rebounds
##           1.834676           1.843552           1.877814
##           steals           blocks           personal_fouls
##           1.329903           1.249977           1.927697
## personal_fouls_drawn
##           1.447514
```

```
#offensive model
```

```
off_mod <- lm( wins ~ field_goal_percentage + three_point_percentage +
               free_throw_percentage + assists + offensive_rebounds + turnovers,
               data = nba_train
             )
```

```
summary(off_mod)
```

```
##
## Call:
## lm(formula = wins ~ field_goal_percentage + three_point_percentage +
##     free_throw_percentage + assists + offensive_rebounds + turnovers,
##     data = nba_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -24.4026  -6.3191   0.2612   6.0136  26.3503
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.768e+02  1.918e+01  -9.218 < 2e-16 ***
## field_goal_percentage  3.442e+00  3.582e-01   9.609 < 2e-16 ***
## three_point_percentage  1.785e+00  2.686e-01   6.647 7.92e-11 ***
```

```
## free_throw_percentage -8.676e-02 1.473e-01 -0.589 0.556
## assists 2.337e-04 2.472e-03 0.095 0.925
## offensive_rebounds 3.506e-02 4.138e-03 8.473 2.77e-16 ***
## turnovers -2.413e-02 4.292e-03 -5.622 3.16e-08 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.06 on 494 degrees of freedom
## Multiple R-squared: 0.4298, Adjusted R-squared: 0.4228
## F-statistic: 62.05 on 6 and 494 DF, p-value: < 2.2e-16

#defensive model
def_mod <- lm( wins ~ steals + blocks + defensive_rebounds + personal_fouls,
              data = nba_train
            )

summary(def_mod)

##
## Call:
## lm(formula = wins ~ steals + blocks + defensive_rebounds + personal_fouls,
##     data = nba_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.130  -7.348   1.026   7.673  24.167
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -19.857518   7.890385  -2.517  0.01216 *
## steals         0.031871   0.007262   4.389 1.40e-05 ***
## blocks        0.034915   0.007657   4.560 6.46e-06 ***
## defensive_rebounds 0.016742  0.002487   6.732 4.64e-11 ***
## personal_fouls -0.009316  0.003005  -3.100 0.00204 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.74 on 496 degrees of freedom
## Multiple R-squared: 0.1957, Adjusted R-squared: 0.1892
## F-statistic: 30.17 on 4 and 496 DF, p-value: < 2.2e-16

# Offensive Interaction FG% × Assists

off_int <- lm( wins ~ field_goal_percentage * assists +
              three_point_percentage + free_throw_percentage +
              offensive_rebounds + turnovers,
              data = nba_train
            )
anova(off_mod, off_int)

## Analysis of Variance Table
##
## Model 1: wins ~ field_goal_percentage + three_point_percentage + free_throw_percentage +
##      assists + offensive_rebounds + turnovers
## Model 2: wins ~ field_goal_percentage * assists + three_point_percentage +
```

```
##      free_throw_percentage + offensive_rebounds + turnovers
## Res.Df  RSS Df Sum of Sq      F Pr(>F)
## 1      494 40553
## 2      493 40452  1      100.9 1.2297 0.268

#Defensive interaction block x de_rebounds

def_int <- lm( wins ~ blocks * defensive_rebounds + steals + personal_fouls,
              data = nba_train
            )

anova(def_mod, def_int)

## Analysis of Variance Table
##
## Model 1: wins ~ steals + blocks + defensive_rebounds + personal_fouls
## Model 2: wins ~ blocks * defensive_rebounds + steals + personal_fouls
## Res.Df  RSS Df Sum of Sq      F Pr(>F)
## 1      496 57200
## 2      495 57183  1      16.467 0.1425 0.7059

#final models

final_offensive_model <- lm(wins ~ field_goal_percentage +
                           three_point_percentage +
                           offensive_rebounds + turnovers,
                           data = nba_train
                         )

summary(final_offensive_model)

##
## Call:
## lm(formula = wins ~ field_goal_percentage + three_point_percentage +
##      offensive_rebounds + turnovers, data = nba_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -24.6000  -6.2752   0.3691   6.0954  26.2831
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.836e+02  1.438e+01 -12.766 < 2e-16 ***
## field_goal_percentage  3.444e+00  3.072e-01  11.212 < 2e-16 ***
## three_point_percentage 1.772e+00  2.672e-01   6.633 8.62e-11 ***
## offensive_rebounds    3.533e-02  4.063e-03   8.695 < 2e-16 ***
## turnovers        -2.354e-02  4.114e-03  -5.723 1.82e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.045 on 496 degrees of freedom
## Multiple R-squared:  0.4294, Adjusted R-squared:  0.4248
## F-statistic: 93.3 on 4 and 496 DF, p-value: < 2.2e-16

final_defensive_model <- lm(wins ~ steals + personal_fouls + blocks + defensive_rebounds,
                           data = nba_train
                         )
```

```

    )

summary(final_defensive_model)

##
## Call:
## lm(formula = wins ~ steals + personal_fouls + blocks + defensive_rebounds,
##     data = nba_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.130  -7.348   1.026   7.673  24.167
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -19.857518   7.890385  -2.517  0.01216 *
## steals           0.031871   0.007262   4.389 1.40e-05 ***
## personal_fouls  -0.009316   0.003005  -3.100  0.00204 **
## blocks           0.034915   0.007657   4.560 6.46e-06 ***
## defensive_rebounds 0.016742   0.002487   6.732 4.64e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.74 on 496 degrees of freedom
## Multiple R-squared:  0.1957, Adjusted R-squared:  0.1892
## F-statistic: 30.17 on 4 and 496 DF,  p-value: < 2.2e-16

#residual analysis

par(mfrow = c(2, 2))

plot(fitted(final_offensive_model),
     resid(final_offensive_model),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted (Offensive)")
abline(h = 0, col = "red")

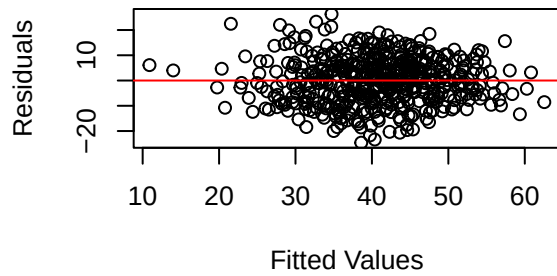
qqnorm(resid(final_offensive_model), main = "Q-Q Plot (Offensive)")
qqline(resid(final_offensive_model), col = "red")

plot(fitted(final_defensive_model),
     resid(final_defensive_model),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted (Defensive)")
abline(h = 0, col = "red")

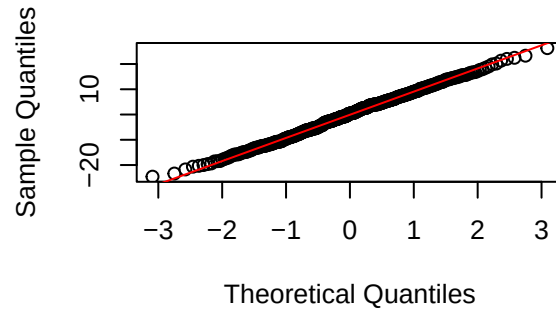
qqnorm(resid(final_defensive_model), main = "Q-Q Plot (Defensive)")
qqline(resid(final_defensive_model), col = "red")

```

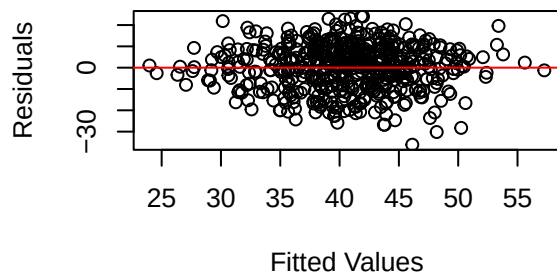
Residuals vs Fitted (Offensive)



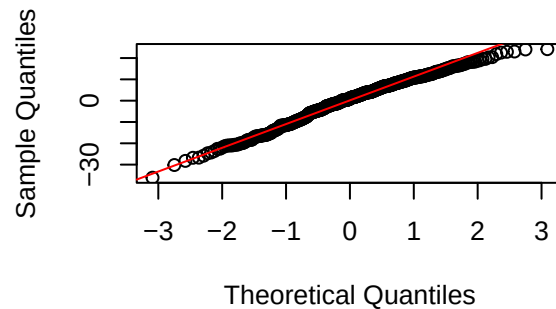
Q-Q Plot (Offensive)



Residuals vs Fitted (Defensive)



Q-Q Plot (Defensive)



```
par(mfrow = c(1, 1))

nba_train <- as_tibble(nba_train)

# Add predictions to training data
nba_train <- nba_train %>%
  mutate(
    pred_off = predict(final_offensive_model),
    pred_def = predict(final_defensive_model)
  )

# Combine into long format for faceting
plot_data <- nba_train %>%
  dplyr::select(wins, pred_off, pred_def) %>%
  pivot_longer(
    cols = c(pred_off, pred_def),
    names_to = "model",
    values_to = "pred_wins"
  ) %>%
  mutate(
    model = ifelse(model == "pred_off", "Offensive Model", "Defensive Model")
  )

# Plot
ggplot(plot_data, aes(x = pred_wins, y = wins)) +
  geom_point(alpha = 0.6) +
  geom_abline(slope = 1, intercept = 0, color = "red", size = 1) +
  facet_wrap(~ model) +
  labs(
```

```

title = "Figure 1: Predicted vs Actual Wins",
x = "Predicted Wins",
y = "Actual Wins"
) +
theme_minimal()

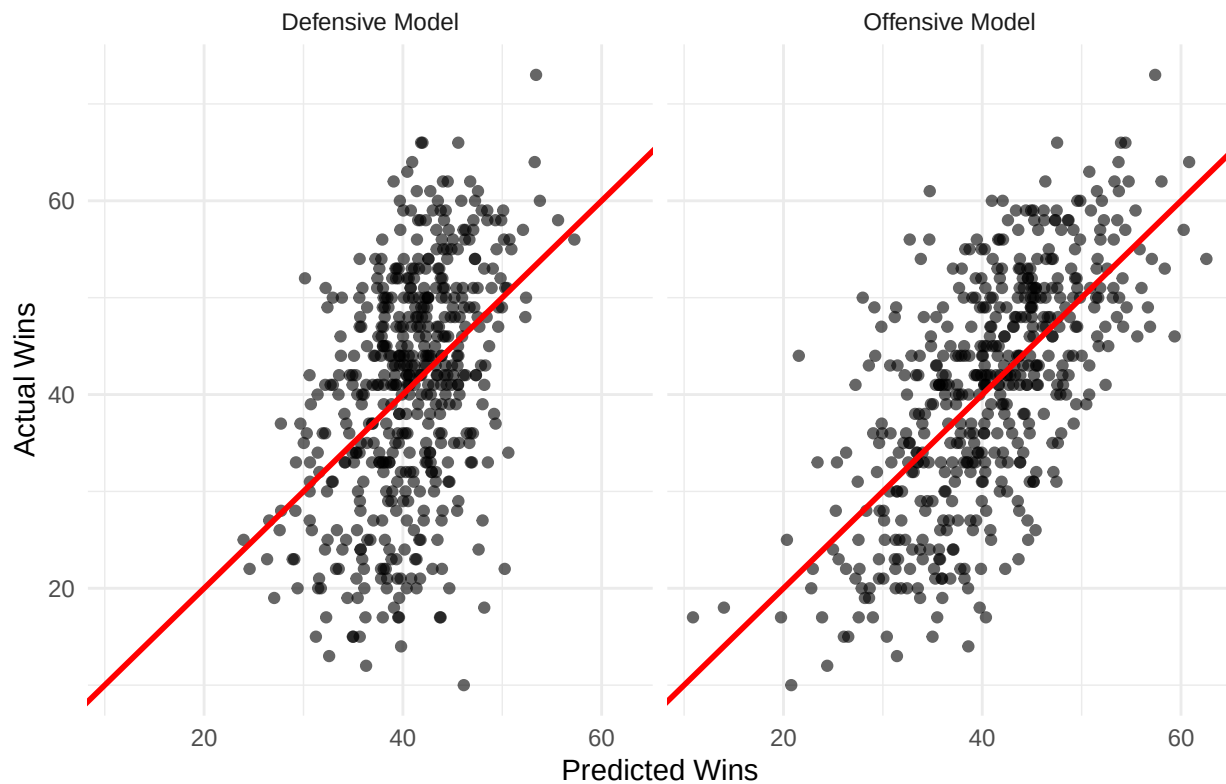
```

```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Figure 1: Predicted vs Actual Wins



```

ggsave("figure1_exploratory.png", width = 6, height = 4, dpi = 300)

```